

VACCINES FROM THE STANDPOINT OF THE PHYSICIAN.¹

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IN resuming this discussion it may be well to remember that its scope is confined to the effects of treatment by vaccines, good, bad, or negative, and, further, that the subjects of prophylaxis and of tuberculin should as far as possible be avoided. In the opening speeches the two most striking statements were the frank admission that the failures of vaccine therapy were probably much more numerous than its successes, and that vaccines when employed should be in addition to, and not as a substitute for, the ordinary methods of treatment. Dr. Phineas Abraham's suggestion that another word should be used in the place of "vaccine" is probably too late, for the American substitute, "bacterin," has not apparently made much headway.

Opinion on vaccine therapy may be divided into the following classes: 1. The optimistic, held by those who, while under the fascination of a new conception of a really rational treatment, have seen several cures in succession. This attitude is usually somewhat modified by time. 2. The pessimistic, more often expressed in private than in print, to the effect that vaccines never do any good and that benefit apparently due to their use is either a coincidence or is perhaps due to suggestion. This latter destructive criticism demands some courage from those who hold it, for it lays them open to the retort that their bad results depend on faulty technique, insufficient experience, or reluctance to give credit where credit is due. I therefore hope that Dr. H. Batty Shaw,⁴ who has so openly expressed his views, will not feel that as secretary of this section of the Royal Society of Medicine it is his duty to smother his convictions. 3. The doubtful or open-minded. This is my own position. Curative vaccine treatment must be admitted to have disappointed the high hopes with which it began, and in this respect its history resembles that of most remedies. The problem of immunity is so complex that it is highly probable that the technique or practice rather than the principle of vaccine therapy is at fault, and that in time this may be so perfected as to establish the position of vaccines as a reliable remedy. The determination of the infecting micro-organism is exposed to well-known fallacies and may be very difficult. Even with all due precautions a wrong vaccine may be given and do harm; the blood of a woman suspected of infective endocarditis was sterile, but agglutinated *Micrococcus melitensis*, and a vaccine was therefore given. Her condition then rapidly deteriorated, and the necropsy, 18 days after the vaccine was begun, revealed infective endocarditis with multiple abscesses in the kidneys. Possibly the conditions of culture on artificial media may so modify the products of the micro-organisms of which the vaccine is composed as to render artificial inoculation far inferior to auto-inoculation. It has therefore been suggested that vaccines should be made from organisms grown on blood or other natural media (Hort²). There is an obvious analogy between vaccine and organo-therapy; the success

of thyroid substitution treatment has been so irrefutable that it has stifled any doubts as to the rationale of the treatment, which might be raised by want of success with other glandular extracts. In Addison's disease, for instance, some link in the chain necessary to secure success in adrenal medication commonly fails to hold; and artificial organo-therapy, like artificial inoculation, shows itself a poor substitute for the normal process. The hypothesis of pluri-glandular insufficiency has brought forth an industry of pluri-glandular therapy which is comparable to that of the "phylacogens."

Further, there is no doubt that vaccine therapy has been used improperly in many ways; thus, it has been employed as a speculation—namely, in cases in which the nature of the infection is assumed but not proved; for example, a stock pneumococcic vaccine is given to a patient thought, from symptoms and signs but not shown bacteriologically, to have pneumococcic infection of the lungs. This attitude, which is quite foreign to its rationale, has damaged vaccine therapy and has exerted an evil influence on medicine generally. The introduction of "phylacogens" recalls the happy-go-lucky polypharmacy of a bygone age. That bad results may follow injections of vaccines must be admitted. I have seen cases in which this sequence of events, and even grave illness, seemed beyond doubt. It may be due to administration of the wrong vaccine, to excessive dose of the appropriate vaccine, or to acquired hypersensitiveness on the part of the patient. Although it may sound paradoxical, the power of vaccines for evil may be regarded as, to some extent, an argument that they may in different circumstances be powerful for good.

At the present time the results of vaccines are so uncertain that their use appears to be justified only when trustworthy therapeutic methods have failed or do not exist. Few, if any, would be prepared to argue that the specific action of vaccines is as certain as that of such drugs as mercury, iodides, bromides, salicylates, digitalis, and arsenic.

That good effects, sometimes dramatic in their intensity, may follow the use of vaccines is also undoubted. But the old question—*post hoc propter hoc* arises. Difficult as it is to decide this problem in the absence of exact controls, it seems wiser to act as if the sequence of events was not a coincidence only, and in cases in which nothing better is available to give our patients the chance of benefit from vaccine treatment. In conclusion, I will refer briefly to two cases of diseases usually resistant to vaccine therapy, in which cure followed this method of treatment.

In the case of a woman with an old mitral lesion an oscillating temperature appeared and a blood culture by Dr. E. L. Hunt showed a micrococcus differing both from the streptococci and pneumococci usually found in malignant endocarditis. The administration of a vaccine was gradually followed by a normal temperature and apparent cure. This case is mentioned because the usual experience is that vaccine therapy fails in infective endocarditis; thus Wynn,⁵ who was successful in other systemic infections, did not obtain any good result in 15 cases of streptococcic endocarditis, and Horder¹ reported a similar experience in 12 cases. But on the other hand it is quite possible that the improvement in the case just mentioned was independent of vaccine treatment, for Libman³ has shown that bacterial-free periods and recovery may occur spontaneously in subacute bacterial endocarditis. In a case of acute pyelitis due to *Bacillus coli* in an infant

¹ A paper read before the Section of Medicine of the Royal Society of Medicine at the opening of a resumed discussion on Vaccines on Feb. 3rd, 1914.

aged 2 years injection of one-sixth of a million of bacilli brought the temperature down at once in a very striking manner. This appears to be an exceptional result, but I cannot help correlating it with the treatment. In chronic local infections of various kinds, such as arthritis and colitis, in which gradual improvement follows vaccine therapy, it is very hard to decide whether the credit is due to the treatment or to Nature unassisted.

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FRESH ALARMS ON THE INCREASE OF CANCER.

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STATEMENTS emanating from the United States have recently been made public alleging an alarming increase of cancer.¹ Mr. F. L. Hoffmann, who is responsible for these statements,² takes the stand that it is absolutely incredible that the figures he gives—namely, a cancer death-rate of 63 per 100,000 of population in 1900 and of 74·3 in 1911 for the United States and for the large cities of 37·2 per 100,000 of population for the five years ending with 1876, and of 80·5 for the five years ending with 1911—do not represent an actual enormous increase, and he deplores the waste of time spent in questioning an undoubted increase.

In endeavouring to appreciate what importance should be attached to these statements it is necessary to call to mind the history of vital statistics in that country, and also their very backward condition at the present day. From 1850 to 1900 the endeavour was made to collect figures of deaths through the enumerators as part of each decennial general census of population of the United States. This attempt at centralisation was doomed to failure because only immediate registration can be effective, and the method was abandoned in 1910. Centralisation of the recording of births, deaths, and marriages has not yet been effected in the United States, and recourse must be had to the reports of the several States. These are of very unequal value, and, indeed, are often nearly valueless.

In 1880 the compulsory registration of deaths, as enforced by municipal or State authority, was first introduced at Massachusetts, New Jersey, the district of Columbia, and for some cities where State control did not obtain. Even to-day there is no registration law at all, or, if so, it has only recently been placed on the Statute Book in many States like Alabama, Georgia, Illinois, and South Carolina. The statistics available for Pennsylvania were utterly valueless up to 1906, as were those for Ohio up to 1909, and those for Missouri and Kentucky up to 1911. Several of those named are Eastern States. The Southern States are still entirely without any sort of reliable registration.

At the present time only 22 of the Eastern, Northern, and Western States make any claim whatsoever to enforce the registration of deaths,

and in many of these the law is most imperfectly carried out. For New York City itself Oertel³ has shown that the greatest laxity and inaccuracy obtains, owing to the negligence of the medical profession and the unwillingness of the authorities to enforce the prescribed penalties. In addition, the attempt is made by 38 municipalities, and by those towns of North Carolina with populations of 1000 and upwards, and by the District of Columbia. Thus from 8,538,266 in 1880 the numbers under registration law have been raised to 59,275,977 in 1911. In other words, while imperfect registration of deaths covered only 17 per cent. of the population of the United States in the year 1880, 31·4 in 1890, and 37·9 in 1900, 63 per cent. are so affected to-day. This is a great improvement in the registration of deaths, but 37 per cent. of the population still remains without even imperfect registration of the causes of death.

By 1900 only 11 States were Registration States. Moreover, prior to 1902 no two States and scarcely any towns used precisely the same methods of certifying deaths. In that year the "standard" certificate was introduced. As revised by Act of Congress in April, 1913, it is now being approved or used in most States of the Union, including Alaska, except Alabama, Georgia, Illinois, New Hampshire, New Mexico, Rhode Island, South Carolina, and West Virginia. It is evident that no comparison is possible between one year and another for the United States, or even between different States.

It must be added emphatically that even the States longest under the "compulsory" registration law—e.g., Connecticut and Massachusetts—have not yet succeeded in enforcing registration of all births. The total population of the United States at the Census of 1910 was 90,753,277. For 1911 the aggregate estimated population of the registration area of the United States was 59,275,977, among whom there were 839,284 deaths recorded. In England and Wales for the same year a population of 36,163,833 yielded 527,810 deaths. For the same year the births and the infantile mortality in the registration area of the United States cannot be estimated with any approach to accuracy, but in England and Wales the births numbered 881,138—i.e., more than seven and a half times the deaths of infants under 1 year (114,600). Thus the United States present a marked contrast. No State has complete registration of births. In 1911 in 11 States only did the births exceed the number of children returned as aged 1 year or under. In 1911 in only two States did the births registered equal the number of infants, and only in three States did the births exceed the figures for infants by the small margin of 10 per cent.⁴ The discrepancies in the above figures of births and deaths in relation to population for the United States, as compared with England and Wales, are to be explained largely by the considerations set forth with reference to the imperfect registration of births and deaths; they signify hundreds of thousands of unpunished offences against the registration laws of the United States in one year. The offences against the Registration Acts in England and Wales amounted in the same year to the small total of 19, all of the offenders being prosecuted and convicted.

These facts are of themselves a sufficient commentary on the very imperfect extent to which

¹ THE LANCET, Jan. 3rd, 1914, p. 47; The Times, Jan. 20th, 1914.

² Frederick L. Hoffmann: The Menace of Cancer, Transactions of the American Gynaecological Society, 1913. Clinical Congress of Surgeons of North America, Chicago, 1913.

³ Oertel: American Underwriter, May, 1913.

⁴ Exclusive of cities of a population of 500,000.